FIBRE PRODUCT CRITERION FOR LIMIT PRESERVING TRANSFORMATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 370-396, following the statement at 350-368.

Editorial note. No omissions; the uniqueness claims are carried by the displayed fibre functor arguments.

Allowed standard prerequisites (editorial).

- Fibre functors of limit preserving transformations. definition-locally-finite-presentation (same chapter, cited at ~375-376). The functor F_{y_i} on (Sch/T_i)_{fppf}.

Source reference. Original definition 049J: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 287–287. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $a : F \to G$ be a transformation of functors $(Sch/S)^{opp}_{fppf} \to Sets$. The following are equivalent*

(1) *$a : F \to G$ is limit preserving, and*
(2) *for every affine scheme T over S which is a limit $T = \lim T_i$ of a directed inverse system of affine schemes T_i over S the diagram of sets*

$$\begin{array}{ccc} \operatorname{colim}_i F(T_i) & \longrightarrow & F(T) \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle a} \\ \operatorname{colim}_i G(T_i) & \longrightarrow & G(T) \end{array}$$

is a fibre product diagram.

Proof. Assume (1). Consider $T = \lim_{i \in I} T_i$ as in (2). Let (y, x_T) be an element of the fibre product $\operatorname{colim}_i G(T_i) \times_{G(T)} F(T)$. Then y comes from $y_i \in G(T_i)$ for some i. Consider the functor F_{y_i} on $(Sch/T_i)_{fppf}$ as in Definition 049J. We see that $x_T \in F_{y_i}(T)$. Moreover $T = \lim_{i' \geq i} T_{i'}$ is a directed system of affine schemes over T_i. Hence (1) implies that x_T the image of a unique element x of $\operatorname{colim}_{i' \geq i} F_{y_i}(T_{i'})$. Thus x is the unique element of $\operatorname{colim} F(T_i)$ which maps to the pair (y, x_T). This proves that (2) holds.

Assume (2). Let T be a scheme and $y_T \in G(T)$. We have to show that F_{y_T} is limit preserving. Let $T' = \lim_{i \in I} T'_i$ be an affine scheme over T which is the directed limit of affine scheme T'_i over T. Let $x_{T'} \in F_{y_T}$. Pick $i \in I$ which is possible as I

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is a directed set. Denote $y_i \in F(T_i')$ the image of $y_{T'}$. Then we see that $(y_i, x_{T'})$ is an element of the fibre product $\mathrm{colim}_i G(T_i') \times_{G(T')} F(T')$. Hence by (2) we get a unique element x of $\mathrm{colim}_i F(T_i')$ mapping to $(y_i, x_{T'})$. It is clear that x defines an element of $\mathrm{colim}_i F_y(T_i')$ mapping to $x_{T'}$ and we win. $\square$